

Hackathon Project Phases Template

Project Title:

Smart Resume Generator

Team Name:

Code creators

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

To generate innovative ideas and features for a Smart Resume Generator that enhances user experience, improves resume effectiveness, and aligns with modern hiring trends.

Key Points:

1. Problem Statement:

- Many job seekers find it difficult to create resumes that effectively highlight their skills and match job requirements.
- Existing resume-building methods lack personalization, keyword optimization, and ATS compatibility, reducing hiring chances..

2. **Proposed Solution:**

- **Automated Formatting & Customization** – Provide professionally designed, customizable templates with real-time formatting assistance.
- **Instant Feedback & Scoring** – Offer real-time suggestions on grammar, structure, and relevance to improve resume effectiveness.
- **Integrated Cover Letter Generator** – Generate personalized cover letters aligned with the user's resume and job description.

3. **Target Users:**

- **Job Seekers** – Recent graduates, professionals, and career changers looking to create optimized resumes for job applications.
- **Students & Interns** – Individuals seeking internships or entry-level positions who need guidance on resume building.
- **Freelancers & Gig Workers** – Independent professionals looking to showcase skills and experience for project-based work.

4. **Expected Outcome:**

- **Enhanced Job Application Success** – Users will have professionally structured, ATS-friendly resumes that increase their chances of getting shortlisted.
 - **Time Efficiency** – Automated suggestions and formatting will reduce the time needed to create and tailor resumes.
 - **Improved Personalization** – AI-driven content recommendations will help users create job-specific resumes with relevant skills and achievements.
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Phase-2: Requirement Analysis

Objective:

Define the functional, technical, and user requirements for the **Smart Resume Generator**.

Key Points:

1. Technical Requirements:

- **Programming Language:** Python, JavaScript, or TypeScript (depending on the stack)
- **Backend:** Node.js with Express or Django/FastAPI (for AI and API services)
- **Frontend:** React.js or Vue.js (for a modern and responsive UI)
- **Database:** PostgreSQL, MongoDB, or Firebase (for storing user data and resumes)

2. Functional Requirements:

- **AI-Powered Resume Generation** – AI suggests optimized content, keywords, and formatting based on job descriptions.
- **ATS Compatibility & Export Options** – Ensures resumes meet ATS standards and allows export in PDF/DOCX formats.
- **Job-Specific Customization** – Users can tailor resumes for different job applications with AI-driven recommendations.

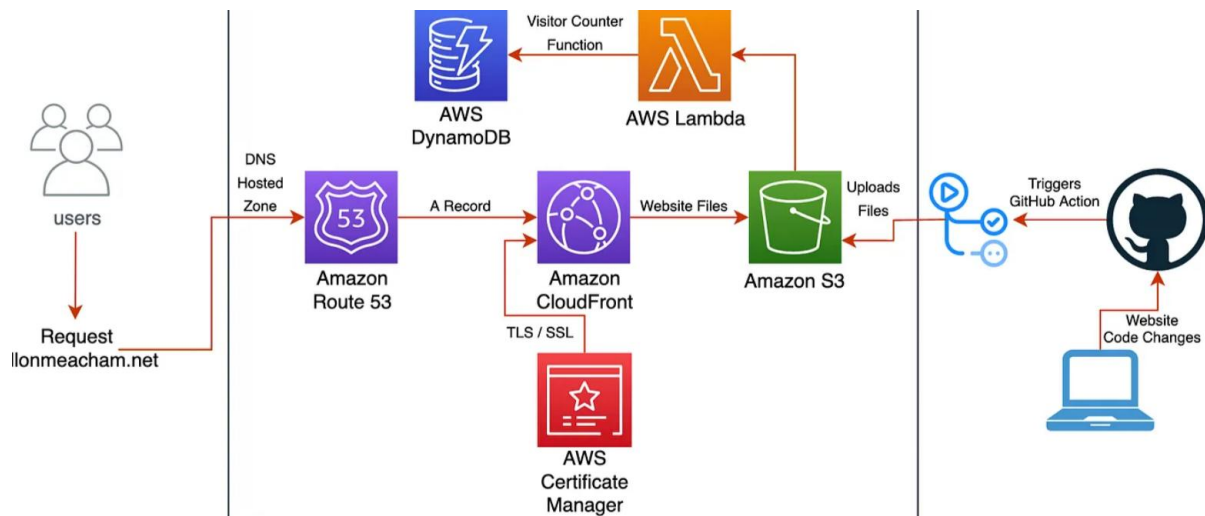
3. Constraints & Challenges:

- **ATS Compatibility** – Ensuring resumes align with different Applicant Tracking System (ATS) requirements.
- **AI Accuracy & Relevance** – Providing precise and contextually appropriate content suggestions.
- **User Experience & Adoption** – Balancing advanced AI features with a simple, intuitive interface.
- **Data Privacy & Security** – Safeguarding user data and complying with privacy regulations.

Phase-3: Project Design

Objective:

To develop a structured, user-friendly, and efficient **Smart Resume Generator** that integrates AI-driven resume optimization, intuitive UI/UX, and seamless functionality.



Key Points:

1. System Architecture:

- **Client Layer (Frontend):** User interface built with React.js or Vue.js for seamless interaction.
- **Application Layer (Backend):** Handles business logic using Node.js with Express or Django/FastAPI.
- **Database Layer:** Stores user data and resumes using PostgreSQL or MongoDB.
- **AI & External Integrations:** Utilizes AI for resume optimization and integrates with job portals & cloud storage.

2. **User Flow:**

- **Input Information:** User enters personal details, work experience, education,
- **Resume Creation:** The system generates a tailored, optimized resume with suggestions for improvement.
- **Review & Export:** User reviews the resume, receives feedback, and exports it in PDF/Word format.

3. **UI/UX Considerations:**

- **Simple Navigation:** Ensure that the navigation is intuitive and easy to follow, minimizing the number of steps required to complete tasks.
- **Inclusive Design:** Make sure your design accommodates all users, including those with disabilities. This includes offering options like keyboard navigation, screen reader support, and sufficient color contrast for visual clarity.

Phase-4: Project Planning (Agile Methodologies)

Objective:

Break down development tasks for efficient completion.

Sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Sprint 1	Requirements Gathering & Planning	High	2 days	Day 2	Product Manager	None	Clear project scope, feature list, and design direction.
Sprint 2	UI/UX Design	High	2 days	Day 4	UX/UI Designer	Sprint 1	User interface designs and flow are finalized.
Sprint 3	Input Form Development	High	2 days	Day 6	Frontend Developer	Sprint 2	Functional input form for

Sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
							collecting user data is implemented.
Sprint 4	Backend Development (Data Processing)	High	2 days	Day 8	Backend Developer	Sprint 3	Backend logic for processing user data and generating resumes.
Sprint 5	Resume Generation Engine	High	2 days	Day 10	Backend Developer	Sprint 4	AI-based engine for generating optimized resumes based on input.
Sprint 6	Export Functionality (PDF/Word)	Medium	2 days	Day 12	Backend Developer	Sprint 5	Export feature for resume in PDF and Word format.
Sprint 7	Testing & Quality Assurance	High	2 days	Day 14	QA	Sprint 6	Perform testing on all features, identify bugs, and improve stability.
Sprint 8	Deployment	High	2 days	Day 16	DevOps	Sprint 7	Application deployed to production, ready for user access.
Sprint 9	Post-Launch Support & Updates	Medium	Ongoing	Day 17 and beyond	All Teams	Sprint 8	Monitor performance, gather user feedback, and push improvements.

Sprint Planning with Priorities

Sprint 1 – Setup & Integration (Day 1)

- (🔴 High Priority) Set up the **environment** & install dependencies.
- (🔴 High Priority) Integrate **Google Gemini API**.
- (🟡 Medium Priority) Build a **basic UI with input fields**.

Sprint 2 – Core Features & Debugging (Day 2)

(🔴 High Priority) Implement **search & comparison functionalities**. (🔴 High Priority) Debug API issues & handle errors in queries. **Sprint 3 – Testing, Enhancements & Submission (Day 2)**

(🟡 Medium Priority) Test API responses, refine UI, & fix UI bugs. (🟢 Low Priority) Final **demo preparation & deployment**.

Phase-5: Project Development

Objective:

The tool aims to streamline the resume creation process by providing smart suggestions, customizable templates, and automated formatting.

Key Points:

1. Technology Stack Used:

- **Frontend:** HTML5, CSS3, JavaScript, React.js or Vue.js, Bootstrap or Tailwind CSS
- **Backend:** Server-Side Development and API Management, Database and User Authentication
- **Programming Language:** Python

2. Development Process:

- Implement **Frontend and Backend Development**
- Develop **AI/ML Integration**
- Optimize **Testing and Debugging**

3. **Challenges & Fixes:**

- **Challenge: User Experience (UX) Design Complexity**

Fix: Conduct thorough user testing and create intuitive UI/UX designs.

- **Challenge: Resume Content Accuracy and Personalization**

Fix: Integrate advanced **AI/NLP** algorithms (like GPT) to offer personalized content suggestions

- **Challenge: Compatibility with Different Formats and Platforms**

Fix: Ensure the tool supports multiple resume formats (PDF, DOCX) and is compatible with various operating systems (Windows, Mac, Mobile).

Phase-6: Functional & Performance Testing

Objective:

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
TC-001	Functional Testing	Generate resume with experience in software development	Resume should display accurate job experience in software development.	☑ Passed	Tester 1
TC-002	Functional Testing	Generate resume with education in	Resume should display educational details	☑ Passed	Tester 2

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
		Business Management	related to Business Management.		
TC-003	Performance Testing	Resume generation time under 3 seconds	Resume should be generated within 3 seconds.	⚠ Needs Optimization	Tester 3
TC-004	Bug Fixes & Improvements	Fix font rendering issues in resume templates	Fonts should render correctly across templates.	☑ Fixed	Developer
TC-005	Final Validation	Ensure resume is downloadable in PDF and DOCX formats	Resume should be downloadable in both PDF and DOCX formats.	✗ Failed - PDF download not working	Tester 2
TC-006	Deployment Testing	Host the application using cloud infrastructure (AWS)	The resume generator should be accessible online without errors.	🚀 Deployed	DevOps

Final Submission

1. **Project Report Based on the templates**
2. **Demo Video (3-5 Minutes)**
3. **GitHub/Code Repository Link**
4. **Presentation**